RENAE GIZZO

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RESEARCH INTERESTS

LiDAR-SAR fusion for forest structure retrieval · Multi-sensor data integration · Machine learning for Earth observation · Scalable environmental data science · Interoperability across Earth observation platforms · Open-source scientific software

EDUCATION

Michigan State University | East Lansing, MI **January 2025 -- Present**
M.S. Department of Plant Biology GPA: 3.9
Thesis: From waveform to backscatter: scalable multimodal deep learning for transferable forest structure mapping from GEDI, NEON Airborne LiDAR, and NISAR

Kent State University | Kent, OH **August 2020 – December 2023**
B.S. Environmental Biology GPA: 3.6
Minor: Geographic Information Science

RESEARCH EXPERIENCE

Ecological Remote Sensing & Modelling Lab | Research Assistant **2025 – Present**
Michigan State Department of Geography
Advisor: Dr. Kyla Dahlin

- Designed a scalable pipeline for integrating diverse Earth observation data (active and passive satellite and airborne sensors) into CNN, random forest, and gradient-boosted models for mapping forest structure across latitudinal gradients
- Refactored legacy lab software (canopyLazR v.2.0) into a maintainable, modular codebase for deriving forest structure metrics from LiDAR point clouds
- Developed components of a multimodal framework integrating LiDAR, hyperspectral, and field observations for 3-D leaf trait mapping
- Led a team of undergraduate technicians on a multi-site field campaign to collect and process foliar samples, leaf spectra, leaf chemistry, and hemispherical photos

Biogeography & Landscape Dynamics Lab | Undergrad Research Assistant **2021 -- 2023**
Kent State Department of Geography
Advisor: Dr. Tim Assal

- Performed dendroecological analysis by cross-dating dead tree cores and building master chronologies using standard dendrochronological methods
- Integrated multispectral vegetation indices, hurricane records, and tree-ring chronologies to conduct superposed epoch analysis of forest responses to hurricane disturbance

Summer Undergraduate Research Experience **2023**
Kent State Department of Geography
Advisor: Dr. Andrew Scholl

- Conducted autonomous and manual UAV surveys across 19 natural lands owned by Kent State University and created orthomosaics, digital surface models, and digital terrain models
- Developed random forest classifiers for converting UAV orthomosaics into land-cover maps

TEACHING EXPERIENCE

Spectral Ecology Summer School | Teaching Assistant **June 2026**

- Provided technical instruction and troubleshooting for M.S. students through early-career faculty using remote sensing software, computational workflows, and field data collection methods to complete capstone research projects

PROFESSIONAL EXPERIENCE

CTL Engineering Inc. | GIS Analyst **January 2024 -- January 2025**

- Administered company GIS database by geocoding legacy project data and leveraging pipelines between ArcGIS Servers and internal network servers for automated data preprocessing and maintenance
- Scripted web mapping applications using the JavaScript API for ArcGIS to provide end users with real-time field data visualization and tracking
- Collaborated with technical staff, project managers, and clients to translate projects into publication quality maps and geospatial data products

HONORS & AWARDS

First place in poster division | AAG East Lakes Division Meeting 2023
Poster title: *"Using Drones for High Resolution Mapping of Kent State University Natural Areas"*

Fellows Research Grant | Environmental Design & Research Institute ($2000) 2023
Project title: *"Using Drones for High Resolution Mapping of Kent State University Natural Areas"*

Fellows Research Grant | Environmental Design & Research Institute ($1000) 2022
Project title: *"Applications of Dendrochronology: When did a living forest become a ghost forest?"*

TECHNICAL SKILLS

Software: Google Earth Engine, QGIS, Esri ArcGIS Suite
Languages: R (proficient), Python (proficient), JavaScript (proficient), C++ (experience)
Machine Learning: PyTorch, scikit-learn, TensorFlow, Keras
Tech Stack: Git/GitHub, HPC/SLURM, Bash/Zsh, VS Code, RStudio

PRESENTATIONS

**=mentee author, †=upcoming*

†**T. Renae Gizzo** & Kyla M. Dahlin "Scalable Multimodal Deep Learning for Transferable Forest Structure Mapping from NEON Airborne Lidar and NISAR" *American Geophysical Union, 2026*

Nayantara Biswas, **T. Renae Gizzo,** Kyla M. Dahlin, Casey Youngflesh "Elevation and vegetation structure drives large-scale intraspecific morphological variation in North American birds" *American Ornithological Society, 2026*

Kyla Dahlin, Meicheng Shen, Alanna Post, **T. Renae Gizzo**, Anthony Bowman, Aaron Kamoske, Zachary Butterfield, Adriana Uscanga, Courtney Ross, Sandhya Sharma, Madelyn Immoos, Scott Stark, Shawn Serbin "An Open-Source Workflow for Building Voxelized Models of Forests with Airborne Lidar and Hyperspectral Data" *International Geoscience and Remote Sensing Symposium, 2026*

*Madelyn Immoos, **T. Renae Gizzo**, Kyla M. Dahlin "Geographic and taxonomic diversity in plant traits within tree canopies" *University Undergraduate Research and Arts Forum, Michigan State University, 2026*

T. Renae Gizzo, & Kyla M. Dahlin, "From waveform to backscatter: linking LiDAR and L-band radar to scale observations of forest structure" *Plant Science Graduate Student Research Symposium, Michigan State University, 2026*

T. Renae Gizzo, Ana Murray, Andrew Scholl & Timothy J. Assal, "Deciphering Atlantic White Cedar Mortality in Southern New Jersey using Dendrochronology" *AAG East Lakes Division Meeting, 2023*

T. Renae Gizzo, Ana Murray & Andrew Scholl, "Using Drones for High Resolution Mapping of Kent State University Natural Areas" *AAG East Lakes Division Meeting, 2023*

T. Renae Gizzo, Ana Murray & Grace Michael, "Applications of Dendrochronology: When did a living forest become a ghost forest?" *Environmental Design and Research Institute Symposium, Kent State University, 2023*

Timothy J. Assal, **T. Renae Gizzo**, Bryanna Norlin, Jennifer Walker "A look back on Hurricane Sandy: measuring the storms effect on a saltwater intolerant species at the forest-marsh ecotone" *Ecological Society of America, 2024*

PROFESSIONAL TRAINING & CERTIFICATIONS

GEDI spaceborne Lidar for Ecosystem Applications | NASA ARSET — November 2025

Wetland Delineation Training Course | Swamp School — March 2024

Part 107 Remote Pilot | Federal Aviation Administration (FAA) — July 2023

SCIENCE OUTREACH

Ecological Climate Project | MSU Science Festival Expo — April 2026, April 2025

- Developed a Raspberry Pi based interactive exhibit using Flask for demonstrating the impacts of forest structure on climate variables

MSU Plant Biology Senior Capstone | Graduate Student Panel Speaker March 2026

MSU Girls in Math & Science Day | Volunteer March 2026

MENTORSHIP

Undergraduate students

Madelyn Immoos | Environmental Biology, Michigan State University 2025 -- 2026

Abigail Snelling | Marine Biology, Michigan State University 2025 -- 2026